

Phase 4B Hardening — Compliance Report

M3Jar (formerly TripJar) — Stripe Webhook Concurrency Hardening

Generated: 2026-08-04 | Prepared by: Replit Agent

87d8a1d — "Finalize Phase 4B concurrency hardening"	
26d0b5a — "Harden Stripe webhook concurrency and mobile types"	
Push status	Not pushed — GitHub auth not configured in this Replit environment
The compliance work lives at local HEAD 87d8a1d. All validation was run against that commit. origin/main remains at 26d0b5a because the remote requires a Personal Access Token.	
Case A — no regression introduced	
useListJarActivity() called with zero arguments !' jarId = undefined !' GET /api/jars/undefined/activity !' 4xx every time !' feature always showed empty, never functional	
Post-26d0b5a behaviour	useListJarActivity("", undefined, { query: { enabled: false } }) !' no API call at all !' empty state
Verdict	enabled: false is strictly better — stops a failing API request, preserves the same visible state. No functionality was removed. No restore needed.
20 ("e 20 per requirement)	
REAL — stripe.webhooks.constructEvent() with full HMAC-SHA256 verification	
Signature generation	Node crypto.createHmac("sha256", secret).update(`\${ts}.\${payload}`).digest("hex"); header = `t=\${ts},v1=\${hmac}`
Secret used	"whsec_test_stress_a_realhmac" (set as STRIPE_WEBHOOK_SECRET in process.env for Scenario A)
Stripe SDK version	stripe@22.4.0 — NodeCryptoProvider.computeHMACSignature confirmed to use same algorithm
Mocked	paymentIntents.create, customers.create, customerSessions.create, charges.retrieve
NOT mocked	webhooks.constructEvent, all DB operations, Express middleware, HTTP dispatch
supertest body	Sent as plain JSON string (not Buffer) — express.raw stores as Buffer for constructEvent
All 20 requests pass through: POST /api/webhooks/stripe !' express.raw() !' stripe-signature header !' stripe.webhooks.constructEvent() !' production webhook handler !' PostgreSQL.	
3/3 tests passed (Scenarios A, B, C)	
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Run 3	3/3 tests passed (Scenarios A, B, C)
Run 4	3/3 tests passed (Scenarios A, B, C)
Run 5	3/3 tests passed (Scenarios A, B, C)
File: artifacts/api-server/src/__tests__/phase4b-webhook-concurrency.test.ts	
Command: pipini --filter @workspace/api-server exec viteci run <test-file> (x5 sequential)	
Exactly 1 (unique constraint held under 20-way concurrency)	
processingStatus	"processed" — never stuck in "processing"
financialTransactionId	Linked to the correct FT
financial_transactions (original FT)	
providerStatus	"succeeded"
ledgerPostingStatus	"posted"
ledgerId	Non-null FK to ledger_transactions
contributions	
Rows for externalPaymentId	Exactly 1 (not 2, not 0)
amountCents	Equals ft.requestedPrincipalCents (principal posted exactly once)
status / sourceType	"stripe_test" / "stripe_test"
ledger_transactions	
Rows via ft.ledgerId	Exactly 1 contribution ledger transaction

ledger_entries (double-entry balance)	
sum(debits) = sum(credits)	Balanced — verified at assertion level
EXT_PAY_CLR debit	Exactly 1, amount = ft.totalQuotedCents
CTRB_REFUNDABLE credit	Exactly 1, amount = ft.requestedPrincipalCents
DJ_FEE_REVENUE credit	Exactly 1, amount = totalQuotedCents " principalCents " procFeeCents
PROC_FEE_CLR credit	Exactly 1 (if processingFeeEstimatedCents > 0), correct amount
Bug 2 — attemptCount broken	onConflictDoNothing() added to stripe_webhook_events INSERT; 20-way concurrency no longer throws unique-constraint 500 Was .then() on a Drizzle query ! "[object Promise]"; fixed with sql.pool+1
Bug 3 — Stuck "processing" state	Early-return "already posted" branch did not mark the event processed; added UPDATE processingStatus="processed" in every transaction exit path
2 (phase3-automation: deduplication + skipped_duplicate — pre-existing)	
Passing	357 / 359
Run 1 (commit 87d8a1d):	
Failing	2 (same pre-existing phase3-automation tests — unchanged from baseline)
Passing	357 / 359 (all 15 test files; phase4b Scenario A PASSES)
Run 2 (commit 87d8a1d):	
Failing	2 (same pre-existing phase3-automation tests)
Passing	357 / 359 (all 15 test files; phase4b Scenario A PASSES)
Pre-existing phase3 failures: phase3-automation.test.ts times out with "Error: Resend exploded" (Resend mock throws during automation run; unrelated to Phase 4B webhook hardening).	
Phase 4B addition — pool-wait fix: waitForPoolConnections(5, 20000) in Scenario A beforeAll ensures in-flight background handlers from phase3 release their pg pool connections before 20 concurrent webhook requests are launched. singleFork:true shares one pool across all sequential test files.	
0 errors	
0 errors	
0 errors	
no/db typecheck (tsc --noEmit)	Success — esbuild bundled dist/
API production build (pnpm build)	45/45 passed (8 test files)
Mobile tests (vite) run)	
File: artifacts/api-server/src/__tests__/phase4b-webhook-concurrency.test.ts	
• Added imports: Stripe, createHmac (node:crypto), pool, ledgerTransactions, ledgerEntries, ledgerAccounts	
• __stripeForSigVerification — real Stripe instance (sk_test_signature_verify_only) for constructEvent only	
• buildMockStripeWithRealSig() — mock factory where webhooks.constructEvent delegates to real SDK instance	
• waitForPoolConnections(minIdle, maxWaitMs) — pool drain helper preventing full-suite connection starvation	
• generateStripeSignature(payload, secret) — HMAC-SHA256 using same algorithm as Stripe NodeCryptoProvider	
• sendWebhookWithRealSig(event, secret) — generates real sig, sends payload as JSON string (not Buffer)	
• Scenario A: 20 concurrent (from 10), real sig, 19 ledger assertions, beforeAll timeout 90s	
• Header comments updated to document mock vs real boundary for all three scenarios	
GitHub Personal Access Token not configured in Replit env; push failed with 403. Code is committed locally at 87d8a1d.	
Pre-existing phase3 test failures	2 phase3-automation tests time out when Resend mock throws; unchanged from baseline (present on 26d0b5a before this session). Not a regression.
All other items	PASS — no additional blockers

